

Concept Attractors in LLMs and their Applications

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Abstract

Large language models (LLMs) often map semantically related prompts to similar internal representations at specific layers, even when their surface forms differ widely. We show that this behavior can be explained through Iterated Function Systems (IFS), where layers act as contractive mappings toward concept-specific Attractors. We leverage this insight and develop simple, training-free methods that operate directly on these Attractors to solve a wide range of practical tasks, including **language translation**, **hallucination reduction**, **guardrailing**, and **synthetic data generation**. Despite their simplicity, these Attractor-based interventions match or exceed specialized baselines, offering an efficient alternative to heavy fine-tuning, generalizable in scenarios where baselines underperform.

1 Introduction

Consider three distinct concepts: the Lord of the Rings universe, the Python programming language, and 19th-century romantic literature. When prompts from these concepts are given to a large language model (LLM) such as Llama 3.1 [2], we see an interesting phenomenon. For each concept, despite lexical variations among its prompts, their intermediate representations appear to collapse to distinct regions at *specific layers* – at which layer this happens varies based on the concept. For instance, prompts such as “Who is Gandalf the Grey?” and “What is the significance of Mount Doom?” share minimal similarity on the surface, yet their representations converge to nearly identical locations at layer 24. We see a similar behavior for Python-related queries such as “Help me implement a binary search tree in Python” versus “How can I find the longest non-repeating substring in Python?” and for prompts for the same genre in literature: “Discuss themes in Pride and Prejudice” and “Any easy way to recognize Byron’s poetry?”. Such a semantic collapse has been reported in some recent results. For instance, [3] notes that transformer models develop a structured latent representations that encode *belief states*. Separately, [4] suggests that due to the internal dynamics of the model, representations converge to “stable” configurations. From a more practical perspective, [5, 6, 7] showed that transformers and LLMs shape their latent space according to the underlying task. These findings, while restricted to smaller models and/or for specific contexts, cumulatively support the idea of representation collapse.

A natural question is whether this concept-specific collapse is implied as a property of some underlying dynamical system already studied in the literature, and if so, what guidance can these existing results

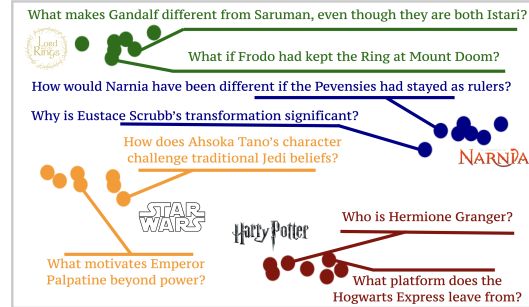


Figure 1: A t-sne[1] plot of the latent representations of Llama3.1-8B for $7 \times 4 = 28$ different prompts, seven each, for the Lord of the Rings universe, Narnia, Star Wars, and Harry Potter. Although the prompts explore different aspects of the universes and share almost no common keywords, we observe a clear clustering based on the different worlds.

provide? Specifically, can we obtain strategies for important downstream use-cases? If $p_1, \dots, p_n$ are a set of prompts related to a specific concept $\mathcal{C}$, we conjecture that the layers of our model may be acting like a dynamical system that maps semantically related inputs to proximal regions, regardless of their form at the “surface”. In other words, the full sequence of layers (leading up to where the representations collapse), if viewed as a unit, implements an iterative (contractive) mapping process to an *Attractor set*, one for each concept. We will see shortly that – to the extent that our hypothesis holds – how existing results are consistent with this view of the collapse phenomena.

Contributions. We show that viewing the LLMs through the lens of Iterated Function Systems [8, 9] offers a meaningful (or at worst, plausible) explanation for both the layer-specific concept clustering and the subsequent generative process. The main practical benefit is that for a wide-variety of downstream tasks, which are often handled piecemeal in the literature, we can obtain a generic scheme that operates under the assumption that operating with the Attractors alone is *sufficient*. We demonstrate that careful interventions on Attractors can provide us lightweight, *training-free* solutions to a wide array of problems, from **programming language translation** and **guardrailing**, to **hallucination reduction** and **synthetic data generation**. Despite the simplicity as well as limited data/compute needs, these solutions turn out to be comparable to existing specialized approaches. Our experiments focus on Llama3.1 8B [2]. However, we see a similar behavior on other LLM families too (in particular, Gemma [10] and Qwen [11]), but avoid an exhaustive analysis of all LLMs.

2 Iterated Function Systems and LLMs

There is mounting evidence that large language models (LLMs) possess emergent capabilities beyond simple rote memorization and statistical pattern matching [12]. Among the many phenomena observed in these models – from in-context learning [13] to compositional reasoning [14, 15] – we focus on a particular representation-convergence property. Our scope is specifically the collapse phenomena at *specific* intermediate layers. To understand this behavior through the lens of dynamical systems, we hypothesize that LLMs implicitly implement a collection of Iterated Function Systems (IFS) during forward propagation through the layers (Fig. 2).

2.1 LLMs implement Iterated Function Systems?

Empirically, we see that for prompts p_i, p_j in each concept $\mathcal{C}$, there exists a layer l where:

$$\lim_{l \rightarrow l_c} \frac{1}{n^2} \sum_{i,j=1}^n |h_l(p_i) - h_l(p_j)| \ll \frac{1}{n^2} \sum_{i,j=1}^n |h_0(p_i) - h_0(p_j)| \quad (1)$$

with h_l denoting the implicit transformation by the LLM up to layer l . This “squashing” of inter-prompt distances suggests that a contractive mapping process is taking place through the layers. Our hypothesis is that this can be understood via the framework of Iterated Function Systems (IFS) [8, 9].

An IFS is defined as a finite set of contractive mappings on a complete metric space. The collective action of these mappings, defined by the Hutchinson operator [9] is:

$$\mathcal{F}(\mathbf{S}) = \bigcup_{i=1}^N f_i(\mathbf{S}) \quad (2)$$

and induces a compact invariant set i.e., $\mathcal{F}(\mathbf{S}^*) = \mathbf{S}^*$, which is called the Attractor of the IFS. More generally, for any initial non-empty compact set $\mathbf{S}_0 \in \mathbb{X}$, the sequence $\{\mathbf{S}_0, \mathbf{S}_1 := \mathcal{F}(\mathbf{S}_0), \mathbf{S}_2 := \mathcal{F}(\mathbf{S}_1), \dots\}$ converges to $\mathbf{S}^*$ in the Hausdorff metric. More generally, an Attractor in a dynamical system is a closed invariant set toward which trajectories from a wide class of initial conditions evolve asymptotically within its basin of attraction, and may take the form of fixed points, periodic orbits, tori, or other Attractors characterized by sensitive dependence on initial conditions [8].

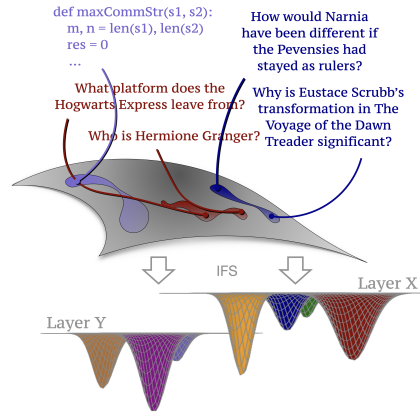


Figure 2: An LLM can be viewed as an IFS that transforms the non-linear manifold of texts into a well-behaving collection of Attractors.

Dynamical systems often exhibit Attractors—sets toward which trajectories converge. Simple systems satisfying Banach’s fixed-point conditions [16] converge to a single point, while others yield more complex structures like limit cycles or strange Attractors [17]. We hypothesize that the iterative application of layer transformations in an LLM induces concept-specific invariant sets—semantic Attractors ($\mathbf{A}_l^{\mathcal{C}}$) for each concept $\mathcal{C}$ —within the latent space at layer l . These compact regions characterize specific concepts, with convergence potentially occurring at different depths depending on the concept.

Once a sequence’s representation enters $\mathbf{A}_l^{\mathcal{C}}$, it is further processed by the remaining layers and output matrix W_{out} to yield a token distribution. Each Attractor may have an invariant measure $\mu_l^{\mathcal{C}}$, describing the distribution of states within it under stochastic dynamics (e.g., varied inputs aligned with concept $\mathcal{C}$). While $\mu_l^{\mathcal{C}}$ is useful for tasks like *synthetic data generation*, it does not directly define next-token probabilities in autoregressive inference, which depend on the specific input-driven state.

The attractors, $\mathbf{A}_l^{\mathcal{C}}$, are linked to the LLM’s operational prefill and decode stages. During prefill, the LLM’s composed layer transformations guide initial representations of an input prompt, $h_0(p)$, towards $\mathbf{A}_l^{\mathcal{C}}$, with the representation $h_l(p)$ landing within this attractor to give the initial semantic context. Then, during decode, each incremental update to the context (by newly generated tokens) is processed by these same underlying layer dynamics. For coherent generation aligned with concept $\mathcal{C}$, the evolving sequence representation at layer l is continually guided towards or kept within the basin of attraction of $\mathbf{A}_l^{\mathcal{C}}$. Thus, $\mathbf{A}_l^{\mathcal{C}}$ acts like a stabilizing latent structure.

Collage theorem. Our operational model takes the transformation performed by the LLM for a concept and approximates it by repeatedly iterating a single affine contractive map [18], $\phi_{\text{eff}} = M_{\text{eff}}V + t_{\text{eff}}$ (with V as a placeholder hidden representation), suggesting that the overall transformation, for a specific concept, can be roughly approximated by an iterated affine dynamics. We want to estimate the parameters (i.e., the matrix M_{eff} and vector t_{eff}) and the number of iterations iter , that best reproduce the observed mapping (Figure 3). This is achieved by minimizing the discrepancy between the LLM’s observed states at the Attractor layer and the states predicted by iterating ϕ_{eff} from the initial prompt representations:

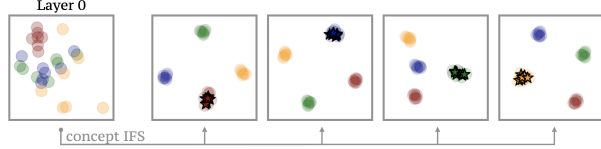


Figure 3: 4 different concepts in layer 0 (before any application of the underlying IFS, and one of the contractions of the underlying IFS we recover by solving the inverse problem for each concept separately. The circles correspond to the true vectors as obtained from the LLM in layer 24 and the stars correspond to the application of the contractions to the points in layer 0.

$$\min_{M_{\text{eff}}, t_{\text{eff}}, \text{iter}} \sum_{j=1}^N \mathcal{D}(h_l(p_j), \phi_{\text{eff}}^{\text{iter}}(h_0(p_j))) \quad (3)$$

subject to M_{eff} being contractive (e.g., its operator norm $|M_{\text{eff}}|_{\text{op}} < 1$). We apply this iter times, and $\mathcal{D}$ is a suitable distance metric. This single map ϕ_{eff} defines a simple Iterated Function System (IFS). The unique Attractor of this 1-map IFS is its fixed point, V^* to which all trajectories $\phi_{\text{eff}}^k(V)$ (for any initial V) converge as k grows. The observed empirical set $\mathbf{A}^{\mathcal{C}}$ is then interpreted as the collection of states reached after iter applications of ϕ_{eff} starting from the initial set S_0 . If, as empirical evidence for many concepts suggests, this 1-map model provides a good first-order approximation, then $\mathbf{A}^{\mathcal{C}}$ would be expected to lie in the vicinity of V^* . The Collage Theorem [8] states that if $\mathbf{A}^{\mathcal{C}}$ is indeed close to the true Attractor V^* of our fitted ϕ_{eff} , then $\mathbf{A}^{\mathcal{C}}$ should be well “collaged” by ϕ_{eff} itself; i.e., $d(\mathbf{A}^{\mathcal{C}}, \phi_{\text{eff}}(\mathbf{A}^{\mathcal{C}}))$ should be small. While the iterated single affine map is simple, for concepts whose empirical Attractors $\mathbf{A}^{\mathcal{C}}$ exhibit more complex geometries (e.g., disjoint sets or intricate fractal structures not well approximated by convergence to a single point), a richer effective IFS comprising multiple affine maps might be necessary. This would involve finding ϕ ’s and an iteration count iter' that minimize $d(\mathbf{A}^{\mathcal{C}}, \mathcal{F}^{\text{iter}'}(S_0))$, where $\mathcal{F}$ is the Hutchinson operator for the candidate set of ϕ ’s. Alternatively, one could model the geometry of $\mathbf{A}^{\mathcal{C}}$ directly by finding an IFS whose intrinsic Attractor matches $\mathbf{A}^{\mathcal{C}}$, by minimizing the collage error. These approaches are more involved but grounded in IFS theory.

Does this perspective add to existing results? Several recent results have indirectly hinted at the IFS-like nature of the LLMs, and more generally transformers, for specific tasks, datasets, and